Revision 12 Post

CS 3244
Machine Learning

NUS | Computing

Recap and Forecast

- Understand the basics of unsupervised learning;
- Understand different unsupervised learning algorithms for:
  - Dimensionality reduction: For aiding visualization, exploratory analysis.
  - Clustering: Finding latent patterns.
- Implement and understand k means clustering;
- Understand the Expectation Maximization (EM) meta-algorithm;
- Apply unsupervised learning on real-world data.

Learning Outcomes for this week:

- Let's get FAT (i.e., Fairness, Accountability and Transparency);
- Have awareness of ML in the larger context, especially how it can affect people's lives other than ourselves;
- Understand the systematic shortcomings of problems with machine learning, in its stages of:
  - Data gathering;
  - Preparation;
  - Model learning.

Epilogue

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

Course Learning Outcomes

1. Understand the basic concepts of machine learning.

2. Apply an appropriate machine learning algorithm for a given problem.

3. Evaluate the performance of a machine learning solution.

4. Use a machine learning tool to carry out machine learning experiments.

Secondary Learning Objectives

In addition to learning about Machine Learning, you have picked up skills that help with your future career.

- **Python** – one of the easiest and more straightforward programming languages to use.
- **Jupyter Notebook / Google Colab** – an intuitive way to mix visualization and code, and get access to state-of-the-art resources.
- **Practical Project** – learn best by applying what you know on a project of your design and communicating it with others.

What we didn't cover

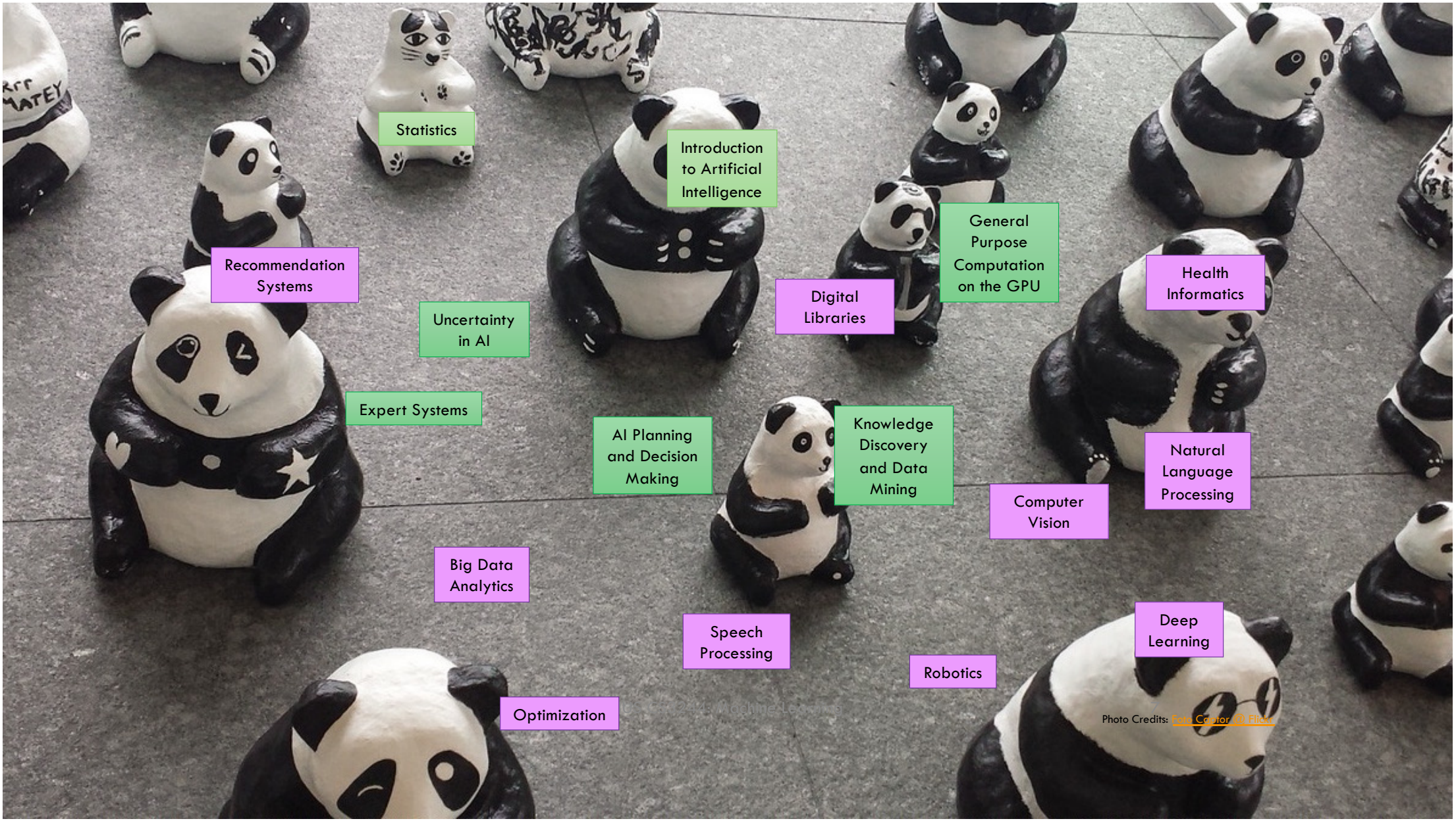
Many things, but notably:

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

Bayesian Models!

But we did touch upon these in tutorials.

You'll see it again in some other NUS machine learning classes.


Statistics
Introduction to Artificial Intelligence
General Purpose Computation on the GPU
Recommendation Systems
Digital Libraries
Health Informatics
Uncertainty in AI
Expert Systems
AI Planning and Decision Making
Knowledge Discovery and Data Mining
Natural Language Processing
Computer Vision
Big Data Analytics
Speech Processing
Deep Learning
Robotics
Optimization
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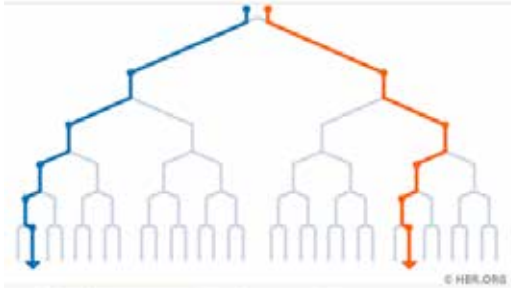
NOW, Everyone needs to know about it.

e Difference Between AI, Machine Learning, and
eep Learning
machine learning, and deep learning are terms that are often used
rchangeably. But they are not the same things.
GS.NVIDIA.COM

Harvard Business Review · October 22, 2015 · Like Page · Sign Up

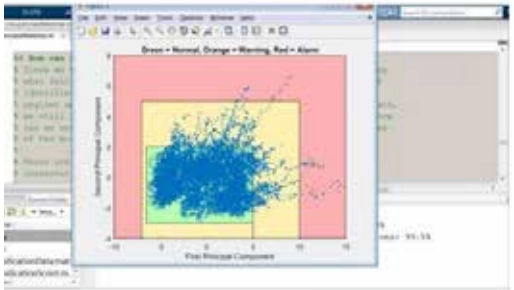
achine learning, explained.



How Machines Learn (And You Win)
Making smart predictions with statistical models
BR.ORG

MATLAB · July 19 at 10:55pm · Like Page · Message

n the fundamentals of machine learning with MATLAB in this video!



achine Learning Made Easy
THWORKS.COM

with the leader of Google Brain.

Machine Learning for Everyone
Recent advances are making machine learning useful outside the tech industry, says the leader of the Google Brain research group.
TECHNOLOGYREVIEW.COM

World Economic Forum · June 18 · Like Page · Sign Up

"You may not know it, but it's all around you."

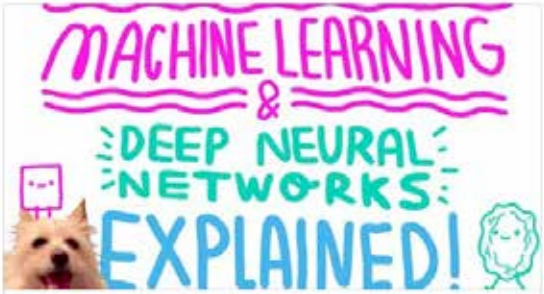
Why you need to understand machine learning
From picking stocks to finding partners, machine learning plays a part in every stage of our lives - yet most of us know little about these powerful tools.
WEFORUM.ORG

The Economist · May 14, 2015 · Like Page · Sign Up

The standard joke about artificial intelligence (AI) is that, like nuclear fusion, it has been the future for more than half a century now. AI is in the news again, for there has been impressive progress in the past few years in a particular subfield of AI called machine learning. But what exactly is that? The Economist explains how machine learning works
http://econ.st/1IA3T6V

Discovery News · September 28, 2015 · Like Page · Watch Video

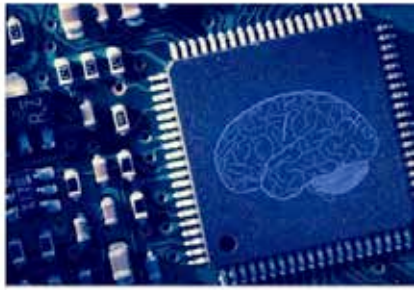
WHAT IS MACHINE LEARNING?: Computers are starting to "think" more and more like humans. What does that mean and how scared should we be? Video: http://dne.ws/1KAducT



What Is Machine Learning?
Computers are starting to
TESTTUBE.COM

Futurism · July 20 at 3:30am · Like Page

Machines can now teach themselves without human inter
how it works—and how it will transform our society.



Understanding Machine Learning [INFOGR
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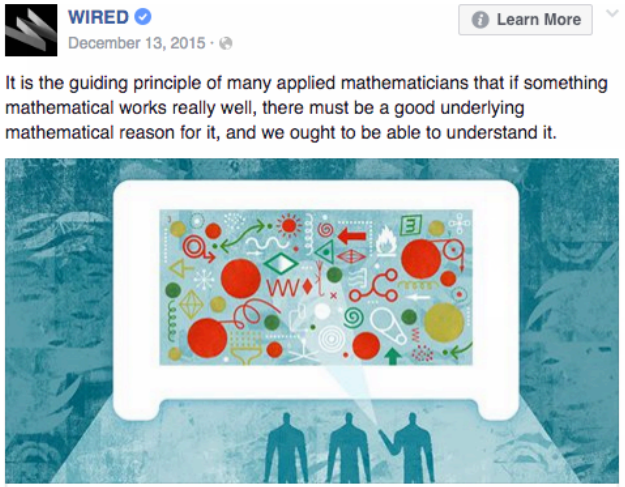
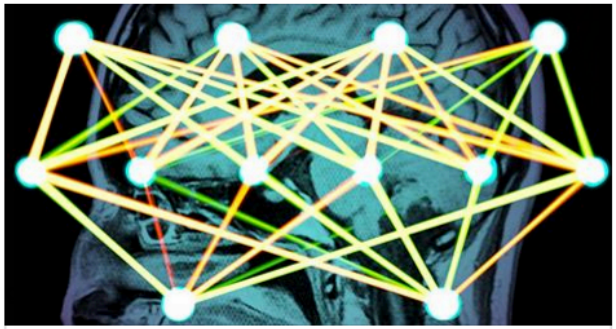
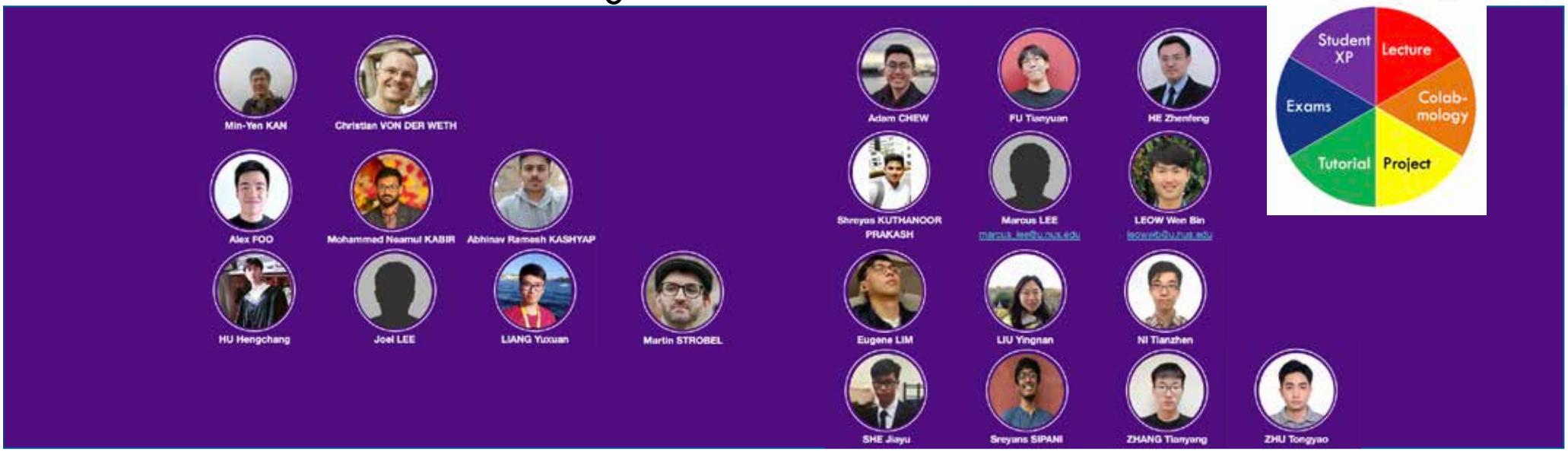
… but perhaps only a few understand it.





Acknowledgments

This course depends on the CS3244 staff subteams that run the course.
We'd like to acknowledge all of their efforts! ~ ~ Let's thank them! ~ ~



Goodbye

Exam 2

9 Nov (Mon) 16:00-18:00.

Venue: Zoom Proctor Rooms, as announced later by 8 Nov (Sun).

Open Web, physical notes and (online) calculator.

Stay in touch with our CS3244 alumni and general interest group on Facebook (search for "cs3244")

Thanks for joining us for the journey!

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Exam 2 Format

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

Exam 2 Format

Same as Exam 1: 25 mins / 25 marks each part × 3 parts.
5 min break between parts

First 2 parts: MCQ / MRQ;
Final part: Short answer / Essay

There will not be any pulse check questions.

There will be programming and calculation questions. Where needed, we provide suitable function call and library prototypes.

- You may refer back to your Colab notebooks if you like.
- Pseudocode ok!

Exam 2 Topics

All W6–W12 topics covered in pre- and post-lecture (through lecture slides, tutorial or Colab).

To be clear: 3 star (***) Tutorial questions and Bonus materials are not examinable

8 topics / 75 marks ≈10 marks/topic

Some of the previous exam questions are from a different lecturer. Min teaches in Sem 1.

Week 06 14 Sep	Regularization and Validation *T04: Bias, Variance and Overfitting*
Week Recess 21 Sep	
Week 07 28 Sep	Reinforcement Learning and Exam 1
Week 08 5 Oct	Neural Networks *T05: Regularization and Validation*
Week 09 12 Oct	Deep Learning *T06: Reinforcement Learning*
Week 10 19 Oct	Decision Trees and Ensembles *T07: Neural Networks*
Week 11 26 Oct	Unsupervised ML: k Means and Expectation Maximization *T08: Deep Learning*
Week 12 2 Nov	Machine Learning Ethics *T09: Decision Trees and Ensembles*

Self-guided Tutorial

No Tutorial Offered

Revision for Exam 2

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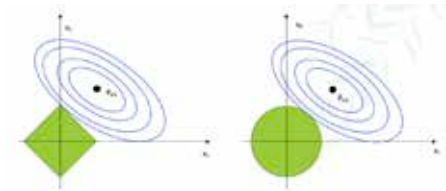
Week 6 – Regularization and Validation

1. Regularization

Give up modeling a subset of of $\mathcal{H}$ to gain better variance.

Minimize $J_{aug}(\theta) = J_{train}(\theta) + \frac{\lambda}{m}\Omega(h)$

Choosing a regularizer

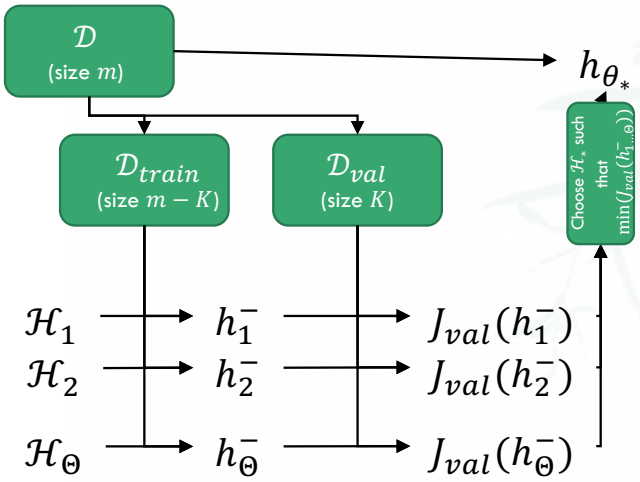


Common $\Omega(h)$: weight decay

Choose λ by validation …

2. (Cross) Validation

$$J_{test}(h) \approx J_{test}(h^-) \approx J_{val}(h^-)$$



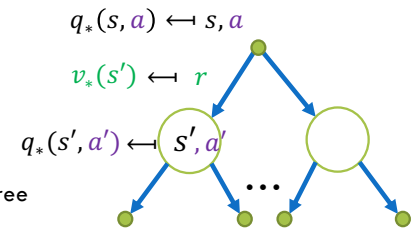
Week 7 – Reinforcement Learning

Understand the basic anatomy and terminology in RL

- Policy, reward, action, observation and (environment, agent) model
- Markov Chain (MC; $\langle \mathcal{S}, \mathcal{P} \rangle$)
- Markov Reward Process (MRP; $\langle \mathcal{S}, \mathcal{P}, \mathcal{R}, \gamma \rangle$)
- Markov Decision Process (MDP; $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}^a, \mathcal{R}, \gamma \rangle$)

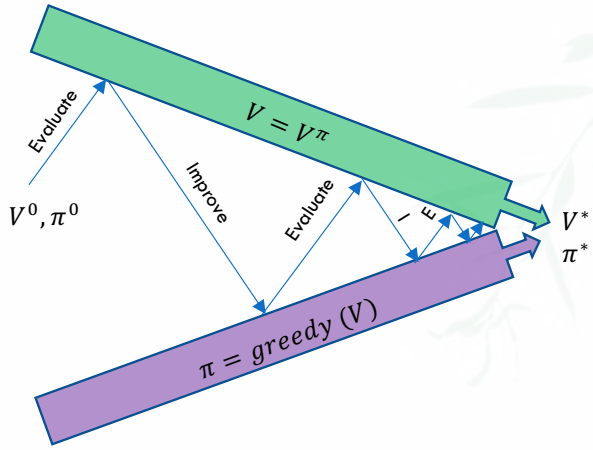
Understand the Bellman equations as a recursive form for calculating future rewards.

- Expectation Equations; linear
- Optimal Equations; max, non-linear
- View as an interleaved tree of states and actions



Solve planning and RL problems using the **Value Iteration** and **Q Learning** algorithms.

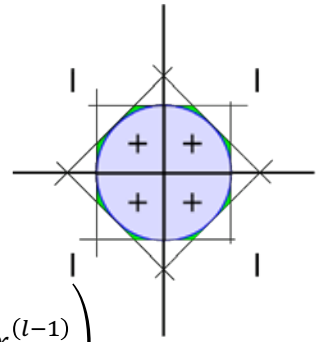
Differentiate them by their modeling assumptions.



Week 8 – Neural Networks

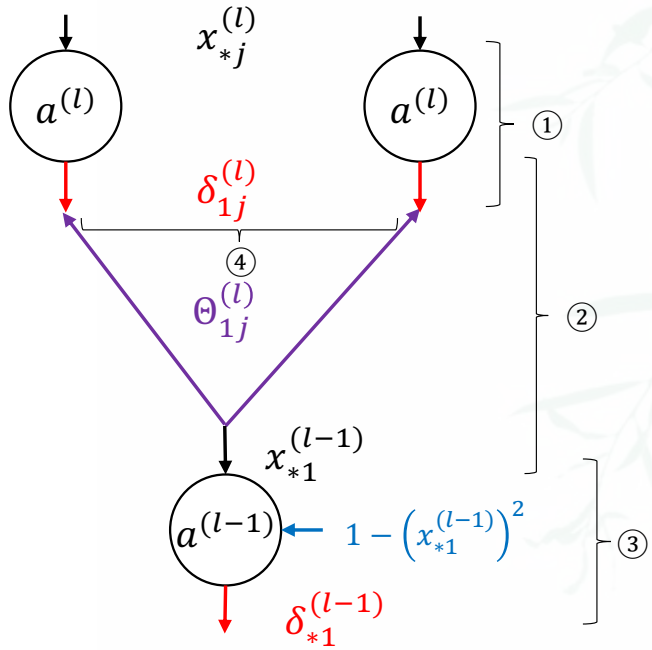
Neural Network

Logical combination
of linear units



$$x_j^{(l)} = g\left(\sum_{i=0}^{d^{(i-1)}} \Theta_{ij}^{(l)} x_i^{(l-1)}\right)$$

where $g(s) = \tanh(s)$



Backprop

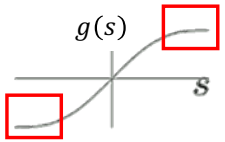
$$\Delta\Theta_{ij}^{(l)} = -\eta\, x_i^{(l-1)} \delta_j^{(l)}$$

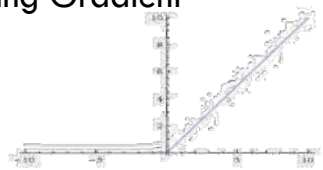
where $\delta_i^{(l-1)} = \left(1-\left(x_i^{(l-1)}\right)^2\right)\sum_{j=1}^{d^{(l)}} \Theta_{ij}^{(d)} \delta_j^{(l)}$

Week 9 – Deep Learning

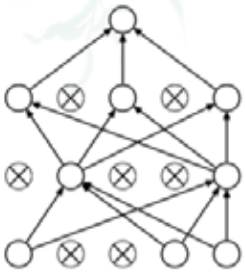
Deep Learning
≡ Stacked NNs
3 important innovations:

1. ReLU – Vanishing Gradient





2. Initialization – Pre-training via autoencoding
3. Dropout – Regularization



(b) After applying dropout.

Two basic variants to regularize toward useful $\mathcal{H}$:

1. <u>Recursive NN</u>: reuse same Θ for multiple time steps in a sequence;
2. <u>Convolutional NN</u>: reuse same Θ for different local regions in an input.

Transfer Learning: Reuse network tuned for a previous task on a current one.

Week 10 – Decision Trees and Ensembling

- Decision Trees

  - Expressive: can overfit to achieve $J_{train} = 0$
  - Need pruning to combat variance

- Bootstrap & Bagging
  - Sample with replacement to generate new $\widetilde{\mathcal{D}}_t$.
  - With each train a h_t, apply uniform voting.
  - Variance lowered through averaging

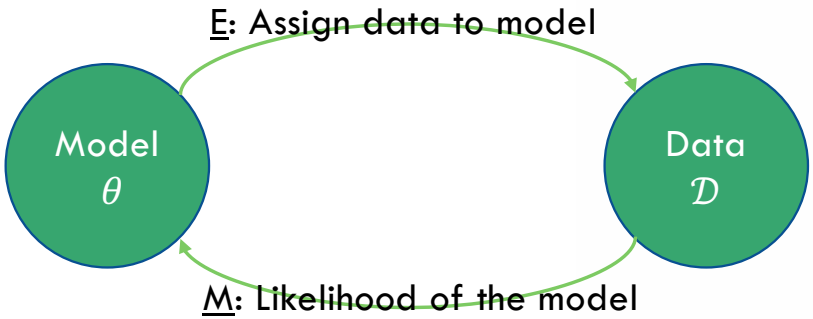
- Boosting
  - Make a strong classifier from simple weak classifiers.
  - Reweight $\widetilde{\mathcal{D}}_t$ to account for easy / hard examples.
  - Lower bias via simple hypotheses.

- Random Forest
  - Combine Bagging with DTL.
  - Introduce more randomness through feature projection.
  - Each h_t tuned to some subset of the features.

Bottom Line: Diversity is good

Week 11 – K means and Expectation Maximization

The EM Algorithm



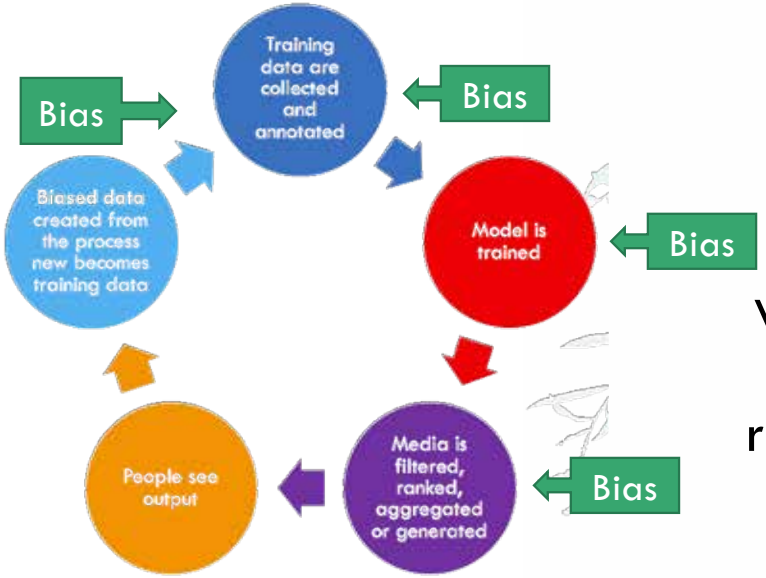
K means as as simplified form of the EM meta-algorithm, where cluster assignments are hard, and using an $\mathcal{L}^2$ norm.

Many instances of EM out there, instantiated for different classes of problems; e.g., "forward-backward", "inside-outside"

Week 12 – Ethics

<u>Bias</u>

- Reporting Bias
- Automation Bias
- Selection Bias
- Group Attribution Bias
- Implicit Bias



<u>Ethics</u>

Remove or control for sensitive fields in training

Validate outcomes are fair with respect to sensitive fields / subpopulations.